Formulas for Gas Math

The formulas behind gas planning, each with a worked example. Depths are in feet of seawater; 33 feet of seawater adds one atmosphere.

Atmospheres absolute (ATA)

> **ATA = (D ÷ 33) + 1**

D is depth in feet of seawater.

Example: At 125 ft: ATA = (125 ÷ 33) + 1 = **4.79 ATA**

Maximum operating depth (MOD)

> **MOD = (PPO_2 ÷ FO_2 × 33) − 33**

PPO_2 is the highest oxygen partial pressure you'll allow. FO_2 is the oxygen fraction, so EAN32 is 0.32.

Example: EAN32 at a PPO_2 of 1.4: MOD = (1.4 ÷ 0.32 × 33) − 33 = **111 ft**

Best mix for a maximum depth

> **Best mix (FO_2) = PPO_2 ÷ ATA**

Use the ATA at your planned maximum depth.

Example: For 125 ft at a PPO_2 of 1.4: 1.4 ÷ 4.79 = 0.292, so **EAN29**

Tank factor (baseline)

> **Tank factor = (rated volume ÷ rated pressure) × 100**

The cubic feet of gas in every 100 psi. For doubles, multiply by 2.

Example: Double LP85s: (85 ÷ 2640) × 100 × 2 = **6.4 cu ft per 100 psi**. Many low-pressure steel tanks reach their nominal volume only with an overfill, so check your tank's spec sheet.

Equivalent narcotic depth (END)

> **END = [(1 − FHe) × (D + 33)] − 33**

The air depth with the same narcotic load. This form treats oxygen as narcotic, as UTD does.

Example: Trimix 17/45 at 240 ft: [(1 - 0.45) × 273] - 33 = **117 ft**

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Equivalent air depth (EAD)

> **$EAD = [(1 - FO_2) \times (D + 33) \div 0.79] - 33$**

The air depth with the same nitrogen partial pressure as your nitrox.

Example: EAN32 at 100 ft: [(0.68 × 133) ÷ 0.79] − 33 = **81 ft**

Partial pressure of a gas

> **Partial pressure = ATA × gas fraction**

Find the ATA first, then multiply.

Example: Oxygen in EAN32 at 100 ft: 4.03 × 0.32 = **1.29 ATA**

Surface air consumption (SAC) rate

> **SAC = (gas used ÷ minutes) ÷ average ATA**

Convert psi used to cubic feet with your tank factor. Measure it on several dives and average.

Example: 1,000 psi from double 104s (tank factor 7.9) is 78.8 cu ft. Over 40 minutes at 70 ft (3.12 ATA): (78.8 ÷ 40) ÷ 3.12 = **0.63 cu ft/min**

CNS oxygen clock

> **CNS % = (minutes at a PPO_2 × 100) ÷ NOAA maximum single exposure**

NOAA's single-exposure limit at a PPO_2 of 1.2 is 210 minutes.

Example: 60 minutes at 1.2: (60 × 100) ÷ 210 = **29%**

For training and planning practice. Check every number with your team, and learn to do this math in a class before you rely on it underwater.

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